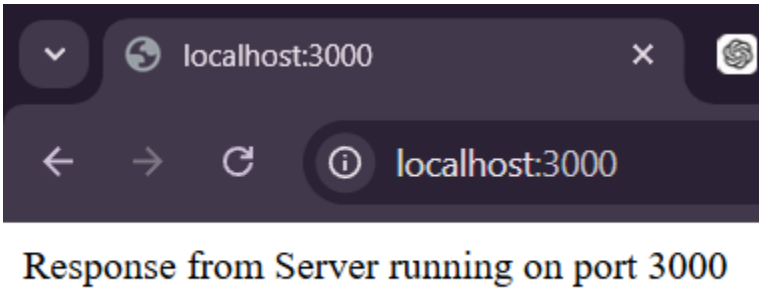
Experiment 6

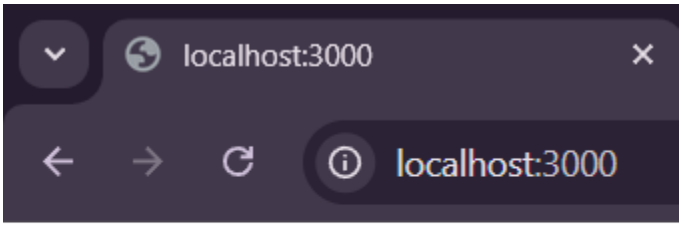
AIM: Set Up NGINX Load Balancing

```
C:\Users\RAJNISH>cd Onedrive
C:\Users\RAJNISH\OneDrive>cd Desktop
C:\Users\RAJNISH\OneDrive\Desktop>cd OST
C:\Users\RAJNISH\OneDrive\Desktop\OST>mkdir load-balance-demo
C:\Users\RAJNISH\OneDrive\Desktop\OST>cd load-balance-demo
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo>npm init -y
Wrote to C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\package.json:
{
  "name": "load-balance-demo",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo>npm install express
added 65 packages, and audited 66 packages in 17s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo>notepad server.js
const express = require("express");
const app = express();
const PORT = process.env.PORT || 3000;
app.get("/", (req,res)=>{
res.send(`Response from Server running on port ${PORT}`);
});
app.listen(PORT,()=>{
console.log(`Server running on port ${PORT}`);
});
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo>node server.js
Server running on port 3000
```

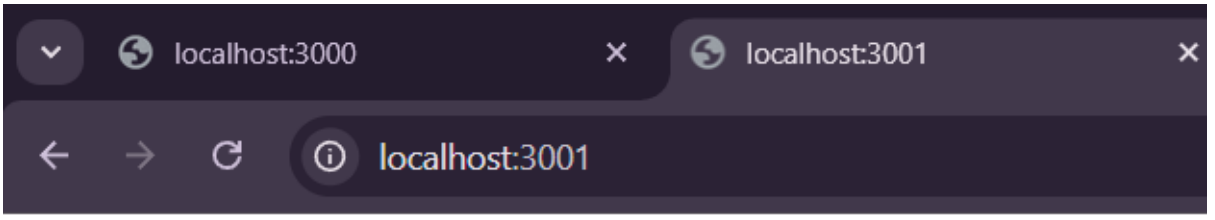



```
C:\Users\RAJNISH>cd OneDrive
C:\Users\RAJNISH\OneDrive>cd Desktop
C:\Users\RAJNISH\OneDrive\Desktop>cd OST
C:\Users\RAJNISH\OneDrive\Desktop\OST>cd load-balance-demo
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo>mkdir load-balancing
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo>cd load-balancing
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing>mkdir server1
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing>mkdir server2
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing>cd server1
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server1>npm init -y
Wrote to C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-
balancing\server1\package.json:
{
  "name": "server1",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server1>npm install express
added 65 packages, and audited 66 packages in 7s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server1>notepad server.js
Wrote to C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-
balancing\server2\package.json:

{:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing>cd server2
  "name": "server2",
```

```
  "version": "1.0.0",rive\Desktop\OST\load-balance-demo\load-balancing\server2>npm init -y
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server2>npm install express
added 65 packages, and audited 66 packages in 4s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server1>notepad server .js
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server1>node server .js
Server 2 running on port 3001
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server2>
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server2>node "server .js"
Server 2 running on port 3001

C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server1>node "server .js"
Server 1 running on port 3001
```

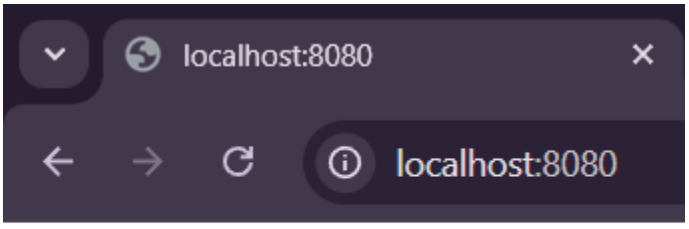


Response from Server 1 (Port 3000)

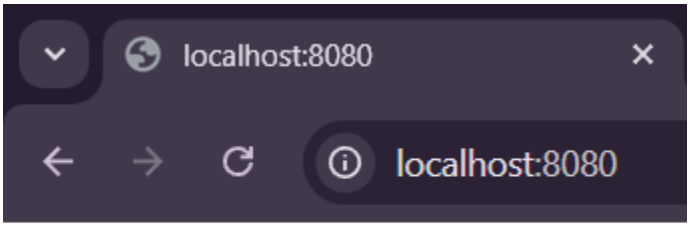


Response from Server 2 (Port 3001)

```
C:\Users\RAJNISH>cd ..
C:\Users>cd ..
C:\>cd nginx-1.29.5
C:\nginx-1.29.5>cd conf
C:\nginx-1.29.5\conf>nginx.conf
worker_processes 1;
events {
   worker_connections 1024;
}
http {
   upstream myapp {

      server localhost:3000;
      server localhost:3001;
   }
   server {
      listen 8080;
      server_name localhost;
      location / {
         proxy_pass http://myapp;
      }
   }
}
C:\nginx-1.29.5\conf>cd ..
C:\nginx-1.29.5>nginx -s stop
C:\nginx-1.29.5>nginx
```

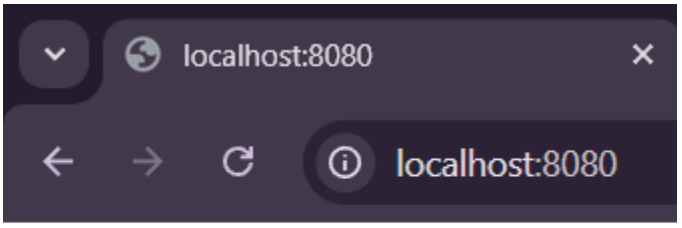


Response from Server 2 (Port 3001)

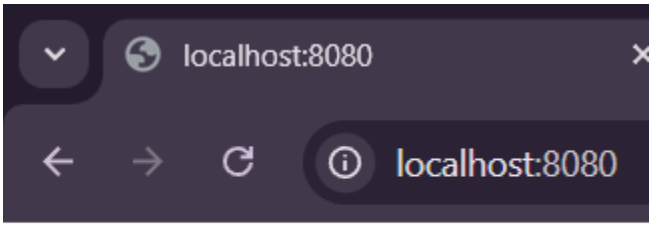


Response from Server 1 (Port 3000)

```
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server1>server.js
const express = require("express");
const app = express();
const PORT = 3000;
app.get("/", (req,res)=>{
setTimeout(()=>{
res.send("Slow Server - Port 3000");
},5000);
});
app.listen(PORT,()=>{
console.log("Server 1 running on port 3000");
});
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server1>node server.js
Server 1 running on port 3000
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server2>notepad "server .js"
const express = require("express");
const app = express();
const PORT = 3001;
app.get("/",(req,res)=>{
res.send("Fast Server - Port 3001");
});

app.listen(PORT,()=>{
console.log("Server 2 running on port 3001");
});
C:\Users\RAJNISH\OneDrive\Desktop\OST\load-balance-demo\load-balancing\server2>node "server .js"
Server 2 running on port 3001
```

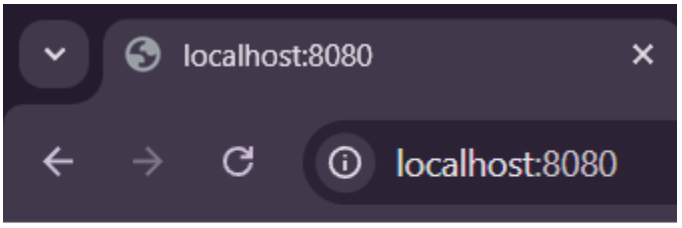


Slow Server - Port 3000

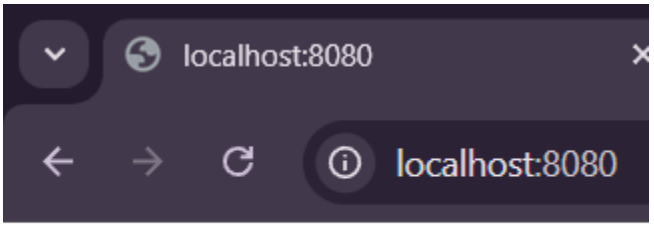


Fast Server - Port 3001

```
C:\Users\RAJNISH>cd ..
C:\Users>cd ..
C:\>cd nginx-1.29.5
C:\nginx-1.29.5>cd conf
C:\nginx-1.29.5\conf>nginx.conf
worker_processes 1;
events {
   worker_connections 1024;
}
http {

upstream myapp {
least_conn;
server localhost:3000;
server localhost:3001;
}
server {
listen 8080;
server_name localhost;
location / {
proxy_pass http://myapp;
}
}
}
C:\nginx-1.29.5\conf>cd ..
C:\nginx-1.29.5>nginx
```



Slow Server - Port 3000



Fast Server - Port 3001